Economics PhD Mathematics and Statistics pre-course syllabus

Course description

The Mathematics and Statistics Pre-course is an eight-week, self-study programme for incoming PhD Economics students. It is aimed at restore or build fluency in the mathematical, probability and statistical tools expected at the start of doctoral study and prepares you for the diagnostic entry exams you will take at the start of your PhD. It is a preparation course, with the aim of identifying and addressing any weaknesses in the key mathematical, probability and statistical tools expected at the start of doctoral study in Economics.

What the course covers

The mathematics part of the course moves from notation, reasoning, algebra and one-variable calculus (`M1`–`M3`) through linear algebra (`M4`–`M5`), multivariable calculus and curvature (`M6`–`M8`), and optimisation (`M9`–`M11`).

The probability and statistics part of the course moves from probability rules and conditional probability (`S1`–`S2`) through random variables, distributions and moments (`S3`–`S5`), and then sampling, estimation and elementary statistical inference (`S6`–`S9`). The sequence ends with confidence intervals and hypothesis tests.

The eight-week schedule gives the exact weekly outcomes, required written readings, resources and what you should read or skip from each source weekly.

How the course works

Each normal week requires approximately 9–11 hours of self-study. This is an estimate and the time you will need will ultimately depend on your prior knowledge and experience. Each week you will:

1. read the assigned written sections;
2. reproduce important worked examples by hand;
3. complete the separate weekly problem sheet when it is released;
4. use progressive hints only after a serious attempt;
5. mark your work from complete solutions;
6. record errors by learning-outcome ID; and
7. reattempt errors two or three days later.

Required work does not include videos, applets, programming or specialist software. Questions are designed to be solved by hand. A basic non-programmable scientific calculator may be permitted for specified assessments, with any necessary critical values supplied.

Assessment and feedback

The course includes two formative checkpoints and two separate timed mock papers: 90 minutes for Maths and 90 minutes for Statistics. They diagnose strengths and revision priorities by learning outcome; they are not pass/fail decisions. The mock papers are practice for the two exams you will take at the start of your PhD.

The required course and every assessment use only outcomes `M1`–`M11` and `S1`–`S9`. Any extension material (marked as `R1` and `R2`) is kept separate from the core course and does not contribute to the weekly core workload or any examination.

First-cohort pilot notice

This is the first time we run this self-study course. All the material is coming from respectable sources freely available online. However, part of the bridge notes, examples and exercises have been created with the help of some LLM models. Therefore there might be some mistakes or inconsistencies in the material. If you find any, please let us know by emailing cristiano.cantore@uniroma1.it with the week, item or source locator. This also means that the stated completion times are provisional planning guidance, not a guarantee, and will be calibrated using your feedback. **For this reason it is important that you record your actual active study time as requested.** Your feedback will improve later versions, and reporting a problem will not disadvantage you.

Checkpoint and mock scores are diagnostic only. They are not used for admission, progression, ranking, funding or grading. Their marking standards and profile bands are provisional for this pilot.

Start here

Read the student syllabus, then use the study protocol and eight-week schedule to plan your first week. The entry-exam information explains how the course assessments relate to the diagnostic examination.

Student syllabus

Learning outcomes by week

Every required weekly topic is attached to an approved learning-outcome ID. Use these IDs in your error log and revision plan.

Week	Required topics and outcome IDs
1	notation, sets, functions, quantifiers and elementary reasoning (`M1`); algebra, powers, logarithms, sums, products and geometric series (`M2`); one-variable calculus, Taylor approximation and elementary optimisation (`M3`); probability rules and counting (`S1`)
2	matrix operations, determinants, inverses, rank and linear systems (`M4`); conditional probability, independence, total probability and Bayes' rule (`S2`)
3	span, linear independence, basis, dimension, eigenvalues, eigenvectors and simple diagonalisation (`M5`); discrete and continuous random variables and required standard distributions (`S3`); expectation, variance, covariance and correlation (`S4`)
4	partial derivatives, column gradients, Jacobians, Hessians and the multivariable chain rule (`M6`)
5	total and implicit differentiation and homogeneous functions (`M7`); quadratic forms, definiteness, convexity, concavity and quasiconcavity (`M8`); joint, marginal and conditional distributions and conditional expectation (`S5`)
6	unconstrained multivariable optimisation and local/global conditions (`M9`); random samples, sampling distributions, sample means and variances, the law of large numbers and the central limit theorem (`S6`)
7	equality-constrained optimisation, multiplier interpretation and the envelope theorem (`M10`); elementary inequality-constrained optimisation and corner solutions under the course convention (`M11`); estimator properties, method of moments and elementary maximum likelihood (`S7`)
8	confidence intervals for a population mean or proportion (`S8`); hypotheses, test statistics, p-values, Type I and Type II errors, power and test–interval links (`S9`)

The exact required readings, section locators and skip instructions are in the eight-week schedule.

Weekly workload

Plan approximately 9–11 hours of core study each week. However, this will ultimately depend on your own background and learning style. A normal allocation is 4–5 hours for reading and reproducing worked examples, 4–5 hours for the weekly problem sheet, and up to one hour for a self-test, marking and error logging. The weekly allocation changes slightly when a checkpoint or one of the mock papers is due.

Course materials

Required materials are written and text-only. The course uses:

- exact section-level external readings;
- course notes and notation reminders;

- worked examples;
- weekly problem sheets;
- progressive hints and complete solutions;
- short self-tests;
- formula or distribution sheets where needed; and
- two checkpoints and two separate mock papers: Maths and Statistics.

No video, applet, programming language, statistical package or specialist software is required. Each week schedule contains specific instructions of what to read and what to skip. **Follow every skip instruction in the schedule, even if the linked source offers additional chapters, exercises or media.**

Study sequence

Use the same sequence each week:

1. Check the week's outcome IDs and retrieve the listed prerequisites.
2. Read only the exact assigned sections and observe every skip instruction.
3. Close the source and reproduce the important definitions, conditions and worked steps from memory.
4. Attempt the weekly problems without opening the full solutions.
5. Consult progressive hints only after a serious attempt.
6. Mark the work, recording the outcome ID and cause of each error.
7. Reattempt missed work two or three days later.
8. Complete the self-test under closed-note conditions.

The study protocol gives a practical session plan and an error-log template.

Assessment points

Point	Timing	Duration	Marks	Main role
Checkpoint A	end of Week 4	60 minutes	30	cumulative formative diagnosis of selected Week 1–4 outcomes
Checkpoint B	end of Week 7	75 minutes	40	cumulative formative diagnosis of selected Week 5–7 outcomes
Maths mock	end of Week 8	90 minutes	55	timed practice across `M1`–`M11`
Statistics mock	end of Week 8	90 minutes	45	timed practice across `S1`–`S9`

All four practice papers use only required outcomes. The two mocks are separate: neither depends on answers or material from the other. Extension material is not examined. See entry-exam information for the approved conditions and the limits of what this course can say about the two entry exam papers.

Working conventions

Use the notation established in the course materials. In particular, vectors and gradients are columns, and the Jacobian of a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ is an m by n matrix. State distribution parameterisations and the conditions needed for a method. For inequality constraints, use the course's maximisation convention with constraints written $g_j(\mathbf{x}) \geq 0$, multipliers $\lambda_j \geq 0$, and Lagrangian term $+\lambda_j g_j(\mathbf{x})$.

What successful completion looks like

By the end of Week 8, you should be able to attempt both 90-minute mock papers under timed conditions, diagnose errors by outcome ID, and identify a short, specific revision list. Completion is evidence of preparation; it is not a guarantee of a particular result in either entry exam paper.

Eight-week schedule

How to use this schedule

This is the complete required reading route for the eight-week core course. Every external reading below identifies an approved resource-map ID and its registered source ID. The four original institutional bridge notes are listed in the relevant weekly route as required point-of-need support. Open only the named sections, follow the skip instruction, and do not treat the surrounding sections of the book or course as assigned.

The time allocations are **provisional planning guidance, not a guarantee**. They include the separate weekly problem sheets that will be authored in a later production phase; this syllabus does not contain exercises. Most core totals stay within approximately 9–11 hours. Week 7 is a 10-hour-35-minute planning estimate because its four exact required source routes are retained. Extension work is outside these totals.

Workload summary

Week	Reading and worked-example reproduction	Weekly problem sheet	Self-test, marking and error log	Timed assessment	Core total
1	4 h 45 min	4 h 15 min	1 h	—	10 h
2	4 h 20 min	4 h	1 h	—	9 h 20 min
3	5 h 40 min	4 h	50 min	—	10 h 30 min
4	4 h	4 h	30 min	Checkpoint A: 1 h	9 h 30 min
5	5 h	4 h	30 min	—	9 h 30 min
6	4 h 15 min	4 h 30 min	30 min	—	9 h 15 min
7	4 h 50 min	4 h	30 min	Checkpoint B: 1 h 15 min	10 h 35 min
8	3 h 30 min	4 h	30 min	Mock papers: 3 h	11 h

For Weeks 4, 7 and 8, the timed assessment sitting is included in the core total. Required post-assessment self-marking, diagnosis and spaced reattempt are recorded separately: no empirical minute allowance is claimed for them, and they are not included in the published core total until first-cohort calibration. They remain required parts of the study cycle.

Required-source reporting route

To report a broken, inaccessible or unavailable required source, email cristiano.cantore@uniroma1.it with the week, item or source locator. You may request this confidentially without disclosing disability or other sensitive personal information. The programme Coordinator will provide an accessible replacement PDF or an approved alternative reading.

Week 1 — notation, foundations and probability rules

Required topics and outcomes

- sets, functions, quantifiers, implications, equivalences, direct reasoning, contradiction and induction (`M1`);
- equations and inequalities, powers, logarithms, functions, sums, products and finite or convergent geometric series (`M2`);
- derivatives, elementary integrals, first- and second-order Taylor approximations and elementary one-variable optimisation (`M3`); and
- events, probability rules and elementary counting (`S1`).

Required readings **Required bridge note.** Before the external sources, read the notation and conventions bridge note. It standardises course notation and is point-of-need support within the existing Week 1 reading and worked-example allocation, not additional workload.

1. `MATH-M1-OSB-REQ / OSBORNE_MM` — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Section 1.1, "Logic: basics and proof by induction" and only the "Intervals" and "Functions" subsections of Section 1.4. **Skip:** in Section 1.4, "Continuous functions" and all later distance and epsilon–delta material; do not use interactive or graphical elements.
2. `MATH-M1-HEF-REQ / HEFFERON_LA` — **Jim Hefferon, *Linear Algebra.*** In the book PDF, read Appendix A-1 "Statements", A-2 "Quantifiers", A-3 to A-4 "Techniques of Proof", and A-5 to A-12 "Sets, Functions, and Relations". **Skip:** all linear-algebra chapters for this assignment and stop after "Sets, Functions, and Relations".
3. `MATH-M2-OPS-REQ / OPENSTAX_PRECALC2E` — **Jay Abramson, *Precalculus 2e.*** Read the assigned parts of Section 4.3, "Logarithmic Functions", Section 4.5, "Logarithmic Properties", Section 11.1, "Sequences and Their Notations", Section 11.3, "Geometric Sequences", and Section 11.4, "Series and Their Notations" through infinite geometric series. In 4.3 use the definition, exponential–log conversions and natural logarithms; in 4.5 use the product, quotient and power rules; in 11.1 use explicit notation only. **Skip:** arithmetic sequences and arithmetic-series formulae, annuities, graphing/calculator/technology exercises, all Media links, online practice tools and applications not needed for the stated algebraic skill.
4. `MATH-M3-OSB-REQ / OSBORNE_MM` — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Section 1.5, "Calculus: one variable", the "One variable" part of Section 5.1 through the interval-maximisation procedure, and the "One variable" part only of Section 5.2. **Skip:** every "Many variables" subsection in 5.1 and 5.2, rotatable graphs, and proofs beyond the stated operational first- and second-order conditions.
5. `MATH-M3-OPS-REQ / OPENSTAX_CALC2` — **Gilbert Strang and Edwin "Jed" Herman, *Calculus Volume 2.*** In Section 6.3, "Taylor and Maclaurin Series", read only "Taylor Polynomials" through the definition, Example 6.11 and Checkpoint 6.10, then Example 6.13 and Checkpoint 6.12. **Skip:** power-series prerequisites and convergence, proofs of Taylor's theorem, general remainder estimates, all `[T]` technology items, graphs, the student project, Taylor-series manipulations and every exercise after 123.
6. `STAT-S1-REQ-MIT / MIT_1805` — **Jeremy Orloff and Jonathan Bloom, MIT OpenCourseWare 18.05.** On the classes and written-materials page, read Class 1 Reading 1b §§2.2–2.4.4 and Class 2 Reading §§2–3. **Skip:** Problem Set 1 Problem 2, which uses a video hint; Problem 3(e, g), which uses R; and all linked videos, interactive questions and applets.

Suggested sequence and output Retrieve basic fraction and exponent rules, complete the readings in the order shown, reproduce at least one example for each outcome, then complete the Week 1 problem sheet and self-test. Finish with an error-log entry for each of `M1`, `M2`, `M3` and `S1`, including "no error yet" where applicable.

Week 2 — linear systems and conditional probability

Required topics and outcomes

- matrix operations, determinants, inverses, rank, elimination and classification of linear systems (`M4`); and
- conditional probability, independence, total probability and Bayes' rule (`S2`).

Prerequisites to retrieve: algebra and functions (`M2`) and probability rules (`S1`).

Required readings

1. `MATH-M4-HEF-REQ / HEFFERON_LA` — **Jim Hefferon, *Linear Algebra.*** In the book PDF, read Chapter One I.1 "Gauss's Method", I.2 "Describing the Solution Set" and III.1 "Gauss–Jordan Reduction"; Chapter

Three IV.1–IV.4 "Matrix Operations"; and Chapter Four I.2 "Properties of Determinants". **Skip:** chapter-end Topics, especially Computer Algebra Systems; the Sage lab and videos; determinant permutation proofs; Cramer's Rule; and all software computation.

2. `STAT-S2-REQ-MIT / MIT_1805` — **Jeremy Orloff and Jonathan Bloom, MIT OpenCourseWare 18.05.** Read Class 3 Reading §§2–7.2. **Skip:** Problem Set 2 Problem 6, which requires R, and all linked videos, interactive questions and applets.

Suggested sequence and output Interleave elimination practice with conditional-probability trees so that both strands receive spaced study. Complete the Week 2 problem sheet, including its cumulative Week 1 retrieval, then self-test and update entries for `M4` and `S2`.

Week 3 — vector spaces, random variables and moments

Required topics and outcomes

- span, linear independence, basis, dimension, eigenvalues, eigenvectors and simple diagonalisation (`M5`);
- PMFs, PDFs, CDFs and the required discrete and continuous distributions (`S3`); and
- expectation, variance, covariance and correlation, including linear transformations and indicators (`S4`).

Prerequisites to retrieve: matrix operations and elimination (`M4`), algebra (`M2`) and basic probability (`S1`).

Required readings

1. `MATH-M5-HEF-REQ / HEFFERON_LA` — **Jim Hefferon, *Linear Algebra.*** In the book PDF, read Chapter Two I.2 "Subspaces and Spanning Sets", II.1 "Definition and Examples" under Linear Independence, III.1 "Basis" and III.2 "Dimension"; then Chapter Five II.2 "Diagonalizability" and II.3 "Eigenvalues and Eigenvectors". **Skip:** complex-root examples, proof-heavy extensions, starred subsections, Jordan form, nilpotence, chapter-end applications, the Sage lab and videos. Use real vectors and matrices only.

2. `STAT-S3-REQ-MIT / MIT_1805` — **Jeremy Orloff and Jonathan Bloom, MIT OpenCourseWare 18.05.** On the classes and written-materials page, read Class 4 Reading 4a §§2.1–3.5, Class 5 Reading 5b §§4–4.5 and Class 5 Reading 5c §§2–5.1. **Skip:** Reading 4a §§3.6–3.7, including the applet; Reading 5c §5.2, which uses R, and §6 on Pareto and other distributions; and Problem Set 3 Problems 2, 5(a–b) and 6(a), which require R.

3. `STAT-S3-REQ-PISHRO-POISSON / PISHRO_NIK` — **Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes.*** Read only the "Poisson Distribution" subsection of §3.1.5. **Skip:** every other §3.1.5 distribution subsection, including the negative-binomial/Pascal and hypergeometric distributions. Treat the Poisson approximation to the binomial as enrichment, not core work.

4. `STAT-S4-REQ-MIT / MIT_1805` — **Jeremy Orloff and Jonathan Bloom, MIT OpenCourseWare 18.05.** On the classes and written-materials page, read Class 4 Reading 4b §§2.1–2.4, Class 5 Reading 5a §§2–4, Class 6 Reading 6a §§3–4.1 and Class 7 Reading 7b §§2–3.1. **Skip:** Class 6 Reading 6a §§5–6; Class 7 Reading 7b §§3.2–4 on the bivariate normal, overlapping uniforms and proofs; and all linked videos or software.

Suggested sequence and output Use three study blocks: vector-space structure, distribution mechanics, and moments/dependence. The registered reading estimates already include time for the assigned source examples. Complete the Week 3 problem sheet and self-test, then update `M5`, `S3` and `S4` in the error log.

Week 4 — multivariable differentiation

Required topics and outcomes

- partial derivatives, column gradients, `m` by `n` Jacobians, Hessians and the multivariable chain rule with dimensionally consistent notation (`M6`).

Prerequisites to retrieve: one-variable differentiation (M3) and matrix dimensions (M4).

Required reading

1. MATH-M6-OSB-REQ / OSBORNE_MM — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read the partial-derivative material only in Section 1.6, "Calculus: many variables", Section 2.1, "Introduction to multivariate calculus" and Section 2.2, "The chain rule". **Skip:** Section 1.6 Exercise 1 and all open/closed-set or topology material, interactive plots, and Section 2.2 Exercises 10–13. Translate the source's derivative-vector notation to the course convention: column gradients and an m by n Jacobian.

Suggested sequence and output Spend the additional example-reproduction time checking dimensions at every step and translating the source convention. Complete the Week 4 problem sheet and self-test, record M6 errors, then sit Checkpoint A under the conditions in entry-exam information.

Week 5 — comparative statics, curvature and joint distributions

Required topics and outcomes

- total and implicit differentiation under stated conditions, homogeneous functions and Euler's theorem (M7);
- quadratic forms and definiteness, convex sets, and convex, concave or quasiconcave functions (M8); and
- joint, marginal and conditional distributions, independence, conditional expectation, iterated expectation and total variance (S5).

Prerequisites to retrieve: linear algebra (M5), multivariable derivatives (M6), conditional probability (S2) and moments (S4).

Required readings **Required bridge note.** Before the conditional-expectation work, read the conditional expectation bridge note. It supplies the course's conditional-expectation conventions within the existing Week 5 reading and worked-example allocation, not additional workload.

1. MATH-M7-OSB-REQ / OSBORNE_MM — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Section 2.3, "Derivatives of functions defined implicitly", Section 2.4, "Differentials and comparative statics" and Section 2.5, "Homogeneous functions". **Skip:** Section 2.3 Exercises 10–12 and Section 2.5 Exercises 14–15. Do not use an implicit derivative without stating the required regularity or nonsingularity condition, and write total differentials with column gradients.
2. MATH-M8-OSB-REQ / OSBORNE_MM — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Section 3.1 on one-variable curvature, Sections 3.2.1, 3.2.2 and 3.2.3 on quadratic forms, Section 3.3 on multivariable curvature, and Section 3.4 on quasiconcavity. **Skip:** rotatable or interactive figures and proof extensions beyond applying the stated tests. Retain only operational set-convexity and curvature conditions, and do not treat quasiconcavity as concavity.
3. STAT-S5-REQ-PISHRO / PISHRO_NIK — **Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes.*** Read the following non-contiguous sections in order:
   - §5.1.1, "Joint Probability Mass Function";
   - §5.1.2, "Joint Cumulative Distribution Function";
   - §5.1.3, "Conditioning and Independence";
   - §5.1.5, "Conditional Expectation";
   - §5.2.1, "Joint Probability Density Function";
   - §5.2.2, "Joint Cumulative Distribution Function"; and
   - §5.2.3, "Conditioning and Independence". **Skip:** §5.1.4, every unassigned part of §5.3 on dependence measures or transformations, MATLAB and linked interactive material.

Suggested sequence and output Move from derivative conditions to curvature, then use the probability strand as a separate block. Reproduce examples inside the registered five-hour reading allocation. Complete the Week 5 problem sheet and self-test, then update M7, M8 and S5.

Week 6 — unconstrained optimisation and sampling

Required topics and outcomes

- candidates, necessary and sufficient conditions, and local versus global solutions in unconstrained multivariable optimisation (M9); and
- independent and identically distributed samples, sample means and variances, sampling distributions, the law of large numbers and the central limit theorem (S6).

Prerequisites to retrieve: gradients and Hessians (M6), curvature (M8), algebra (M2) and distributions/moments (S3–S4).

Required readings

1. MATH-M9-OSB-REQ / OSBORNE_MM — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Section 4.2, "Optimization: definitions", Section 4.3, "Existence of an optimum", and Sections 5.1, 5.2 and 5.3 on necessary conditions and local/global optima. **Skip:** the proof in Section 4.3 and topology beyond the operational statement "continuous on a closed and bounded feasible set", interactive graphs, and one-variable repetition already assigned under M3.
2. STAT-S6-REQ-PISHRO / PISHRO_NIK — **Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes.*** Read the following non-contiguous sections in order:
   - §7.1.1, "Law of Large Numbers";
   - §7.1.2, "Central Limit Theorem";
   - §7.1.3, "Solved Problems", only the registered problems;
   - §8.1.1, "Random Sampling", from "Random Sampling" through "Sample Mean" only; and
   - §8.2.2, "Point Estimators for Mean and Variance", only the definition and elementary sample-variance discussion. **Skip:** in §8.1.1, stop before "Order Statistics"; in §8.2.2, stop after the elementary sample mean and sample variance; do not use simulations, MATLAB or other software.

Suggested sequence and output Reproduce one candidate-classification example and one standardisation/limit example before the problem sheet. Complete the Week 6 self-test and update M9 and S6.

Week 7 — constrained optimisation and estimation

Required topics and outcomes

- equality constraints, Lagrange multipliers, multiplier interpretation and the envelope theorem (M10);
- elementary inequality constraints, complementary slackness and explicit interior/corner checks under the course convention (M11); and
- parameters, estimands, estimators and estimates; bias, variance, mean squared error and consistency; and elementary method-of-moments and maximum-likelihood estimation (S7).

Prerequisites to retrieve: unconstrained optimisation (M9), curvature (M8), sampling (S6) and moments (S4).

Required readings **Required bridge note.** Before the estimator and method-of-moments work, read the estimators and method of moments bridge note. It standardises the course terminology within the existing Week 7 reading and worked-example allocation, not additional workload.

1. MATH-M10-OSB-REQ / OSBORNE_MM — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Sections 6.1.1, 6.1.2, 6.1.3 and 6.1.4; in Section 6.2, read only the statement and one simple example; then read Section 6.3, "The envelope theorem". **Skip:** general proofs in 6.2, advanced multiple-constraint examples and exercises needing results beyond M8–M9. Re-encode constraints and multiplier signs using the course convention before interpretation.
2. MATH-M11-OSB-REQ / OSBORNE_MM — **Martin J. Osborne, *Mathematical Methods for Economic Theory.*** Read Section 7.1 on the Kuhn–Tucker conditions, the sufficiency result and examples only in

Section 7.3, Section 7.4 on non-negativity conditions, and the Section 7.5 summary. **Skip:** Section 7.2 in full, proofs in 7.3 and advanced constraint qualifications. Translate every problem from the source convention to the course convention: maximise with $g_j(\mathbf{x}) \geq 0$, use $\mathcal{L} = f + \sum_j \lambda_j g_j$, and require $\lambda_j \geq 0$.

3. `STAT-S7-REQ-PISHRO / PISHRO_NIK` — **Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*.** Read the definitions of sample, statistic, estimator and estimate in §8.1.1, "Random Sampling", then read the following non-contiguous point-estimation sections in order:
   - §8.2.0, "Point Estimation";
   - §8.2.1, "Evaluating Estimators";
   - §8.2.2, "Point Estimators for Mean and Variance"; and
   - §8.2.3, "Maximum Likelihood Estimation". **Skip:** §8.2.4 on asymptotic properties of maximum-likelihood estimators, §8.5 on linear regression, order statistics and all software-based work.
4. `STAT-S7-REQ-PSU-MOM / PSU_STAT415_MOM` — **Penn State STAT 415, primary course author Laura Simon.** Read Lesson 3 §3.2, "Method of Moments": the definitions, "One Form of the Method", and Examples 3.8 and 3.9 only. **Skip:** "Another Form of the Method" and Examples 3.10 and 3.11, including gamma-distribution material.

Suggested sequence and output Write the course constraint convention at the top of every optimisation session. Keep a separate terminology card for `S7`. Complete the Week 7 problem sheet and self-test, update `M10`, `M11` and `S7`, then sit Checkpoint B.

Week 8 — statistical inference and the two mock papers

Required topics and outcomes

- construction and interpretation of elementary confidence intervals for a population mean or proportion under stated normal, large-sample or Student's t conditions (`S8`); and
- null and alternative hypotheses, one- and two-sided tests, test statistics, p-values, Type I and Type II errors, power and test–interval connections (`S9`).

Prerequisites to retrieve: sampling distributions and limit results (`S6`) and estimator terminology/properties (`S7`).

Required readings Required bridge note. Before selecting or interpreting critical values, read the distributions and critical values bridge note. It provides point-of-need distribution and critical-value support within the existing Week 8 reading and worked-example allocation, not additional workload.

1. `STAT-S8-REQ-PISHRO / PISHRO_NIK` — **Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*.** Read the following non-contiguous sections in order, restricted to intervals for means and proportions under normal, large-sample and Student's t conditions:
   - §8.3.1, "The General Framework of Interval Estimation";
   - §8.3.2, "Finding Interval Estimators";
   - §8.3.3, "Confidence Intervals for Normal Samples"; and
   - §8.3.4, "Solved Problems", restricted to the registered problems. **Skip:** Problem 1, Problem 4 and Problem 6(2), including variance intervals; do not use software.
2. `STAT-S9-REQ-MIT / MIT_1805` — **Jeremy Orloff and Jonathan Bloom, MIT OpenCourseWare 18.05.** On the classes and written-materials page, read Class 17 Reading 17b §§3–7.1; Class 18 Reading §§3.1–5.4 except §5.3; and Class 23 Reading 23a §§4 and 7. **Skip:** Class 18 §§5.3 and 5.5; Problem Set 8 Problems 1(f–g), 3(b)(iii–iv) and 5; and Problem Set 9 Problems 2–6, which enter Bayesian, chi-squared, F, ANOVA or R material.

Suggested sequence and output Build a one-page comparison of confidence-interval and testing decisions, then complete the Week 8 problem sheet and self-test. Update `S8` and `S9`, review the full error log, and sit the

separate 90-minute Maths and 90-minute Statistics mock papers under timed conditions. After marking both, identify no more than five outcome IDs for final targeted revision.

End-of-course extensions

The required sequence ends after Week 8 and after elementary statistical inference. The advanced extensions page is separate from this schedule, it proposes additional material that you should study at some point in the future. However this material will not be examined in the entry exams.

Study protocol

Purpose

Use this protocol every week. It turns a large reading list into repeated, checkable study and makes your error log useful for later revision. The normal core workload is approximately 9–11 hours per week; use the exact allocation in the eight-week schedule.

Before the week starts

1. Write the week's learning-outcome IDs at the top of a new page.
2. Read the prerequisite and retrieval line in the schedule.
3. Reserve three to five study blocks across the week rather than one long sitting.
4. Reserve a second attempt two or three days after the main problem-solving block.
5. Open only the exact reading sections listed in the schedule.

Required work is text-only. You do not need a video, applet, programming language, statistical package or specialist software. If a linked source offers one, follow the schedule's skip instruction.

The standard cycle

1. Read actively For each assigned section:

- identify the definition, method or result attached to the stated outcome;
- record every condition needed for the result;
- note the dimensions, support or parameter restrictions where relevant;
- stop at the assigned locator; and
- obey the explicit skip instruction before following any source navigation.

Do not copy paragraphs. Write a brief original summary after closing the source.

2. Reproduce worked examples Close the worked solution and reproduce the example by hand. Label each step with its purpose: representation, condition, transformation, calculation or interpretation. Reopen the source only after you have reached a result or a specific impasse.

If your result differs, locate the first divergent step. Record that step, not merely the final mismatch.

3. Attempt the weekly problem sheet Start without hints or full solutions. Write conditions and conventions before calculating. If you are stuck, make one specific written statement about what is known and what remains to be found.

After a serious attempt, use progressive hints in order. Stop reading as soon as a hint gives you a next step, then return to your own work. Do not open the full solution until the attempt is complete.

4. Mark and diagnose Mark from the complete solution. Use a different colour or margin, but do not erase the original attempt. Classify the first cause of each error:

- `CONCEPT` — a definition, condition or interpretation was missing;
- `REPRESENTATION` — the event, equation, matrix, function or distribution was set up incorrectly;

- `METHOD` — the wrong procedure or result was selected;
- `ALGEBRA` — the setup was valid but manipulation failed;
- `NOTATION` — dimensions, orientation, symbols, support or parameters were inconsistent;
- `ARITHMETIC` — the mathematical route was correct but calculation failed; or
- `TIME` — you could complete the work accurately but not within the target.

5. Reattempt after spacing Two or three days later, reattempt each incorrect or incomplete part from a blank page. Do not reread your first solution before starting. Close an error only when you can produce and explain a correct solution without the hint.

6. Self-test under closed-note conditions Complete the weekly self-test without notes, sources or solutions. Use only the calculator permission stated for that artifact. Mark it immediately and add any new errors to the same log.

7. Targeted recovery without a progression pause If your score on a current-outcome self-test is below 60%, or an item remains unresolved after its spaced reattempt, take a targeted recovery route before the next retrieval block. Use the relevant week's prerequisite/retrieval line, reproduce the named worked example for that outcome, complete the first matching exercise part, record the result in the error log, then reattempt the original item from a blank page.

This is a study recommendation, not a hard pause. It has no effect on admission, progression, ranking, funding, grading or sanctions.

Error-log template

Use one row per meaningful error or uncertainty.

Date	Outcome ID	Source or item	First cause	Correct principle or method	Reattempt date	Closed?
YYYY-MM-DD	`M4`	example or item ID	`METHOD`	one-sentence correction	YYYY-MM-DD	yes/no

Do not use "careless" as a cause. Name the observable step that failed.

At the end of the week, add a "no error yet" entry for any current outcome that has not produced an error. This prevents an empty log from being mistaken for evidence of mastery.

A workable weekly rhythm

Session	Main activity	Typical duration
1	prerequisite retrieval and first reading block	90–120 minutes
2	remaining reading and example reproduction	120–180 minutes
3	first half of the weekly problem sheet	120–150 minutes
4	second half of the problem sheet and marking	120–150 minutes
5	spaced reattempt, self-test and error-log review	45–75 minutes

Checkpoint and mock weeks use the adjusted totals in the schedule. Reduce unstructured rereading first; do not remove the spaced reattempt or the timed assessment.

When time is short

Protect the core in this order:

1. required reading at the exact locator;
2. one reproduced example for each current outcome;
3. the complete weekly problem sheet;
4. marking and diagnosis;
5. spaced reattempt; and
6. the self-test.

Do not replace required work with extension material. Do not compensate for a missed week by assigning yourself unlisted chapters. Carry unresolved outcome IDs into the next week's retrieval block and use the error log to choose the minimum targeted review.

Source and solution discipline

External sources remain the work of their named authors or organisations. Use the links and exact locators in the schedule; do not redistribute linked material.

Keep hints and solutions closed during first attempts. For checkpoints and the mock, keep them unavailable for the entire timed sitting.

Entry-exam information

The pre-course prepares you for a diagnostic entry examination consisting of two separate papers: Maths and Statistics. The diagnosis is intended to identify preparation needs; this course does not define a pass/fail threshold.

The examination dates, locations, invigilation and final candidate instructions will be announced in due course.

Examined preparation boundary

The examined preparation boundary is exactly the required learning outcomes `M1`–`M11` and `S1`–`S9` in the student syllabus. The statistics boundary ends with elementary estimation, confidence intervals and hypothesis testing (`S7`–`S9`). Extension outcomes have no checkpoint, mock or diagnostic-exam mapping.

Course assessment sequence

The course supplies four practice papers.

Assessment	Placement	Duration	Marks	Required coverage
Checkpoint A	end of Week 4	60 minutes	30	selected outcomes `M1`–`M4`, `M6` and `S1`–`S4`
Checkpoint B	end of Week 7	75 minutes	40	selected outcomes `M8`–`M11` and `S5`–`S7`
Maths mock	end of Week 8	90 minutes	55	every Maths outcome `M1`–`M11`
Statistics mock	end of Week 8	90 minutes	45	every Statistics outcome `S1`–`S9`

The checkpoints are formative and self-marked. Use their marking schemes and diagnostic profiles to identify outcome-level revision priorities. The mock papers are timed practice; their questions are different from the actual examination papers.

During the controlled first-cohort pilot, all checkpoint and mock scores are diagnostic only: they are not used for admission, progression, ranking, funding or grading. Their marking standards are provisional, and model review does not substitute for qualified human marker standardisation before any consequential reuse.

Mock structure

The mock is split into a 90-minute, 55-mark Maths paper and a separate 90-minute, 45-mark Statistics paper. Each begins with a short diagnostic section.

The Maths paper then has longer sections on:

- linear algebra (`M4`–`M5`);
- multivariable calculus (`M6`–`M7`);
- curvature and optimisation (`M8`–`M11`).

The Statistics paper then has longer sections on:

- probability, random variables and conditional distributions (`S1`–`S5`);
- sampling and limit results (`S6`); and
- estimation and inference (`S7`–`S9`).

Every required outcome has at least one explicit scorable mapping on its subject paper. Each paper score should therefore be read together with its diagnostic profile, not on its own.

Examination paper structure

The examination follows the same split: one 90-minute Maths paper and one separate 90-minute Statistics paper. The examination questions are not the mock questions.

Practice conditions

For both checkpoints and each mock paper:

- complete the paper without notes or solutions;
- do not use hints during the timed attempt;
- a basic non-programmable scientific calculator is permitted;
- programming, computer algebra and statistical software are not permitted or required;
- use the supplied formula sheet and any supplied probability table or critical values; and
- show the method, conditions and interpretation needed for the allocated marks.

All questions are intended to be hand-solvable. Calculator use never replaces the derivation or reasoning shown on the page.

Interpreting feedback

Use the mathematics and probability/statistics profiles separately. For each missed part:

1. record its outcome ID;
2. identify the first error using the categories in the study protocol;
3. return to the corresponding week in the eight-week schedule; and
4. reattempt the item after two or three days.

Provisional readiness descriptions in course feedback are diagnostic bands, not entry-exam pass marks. The programme, not this pre-course, determines how the two entry exam papers are used.